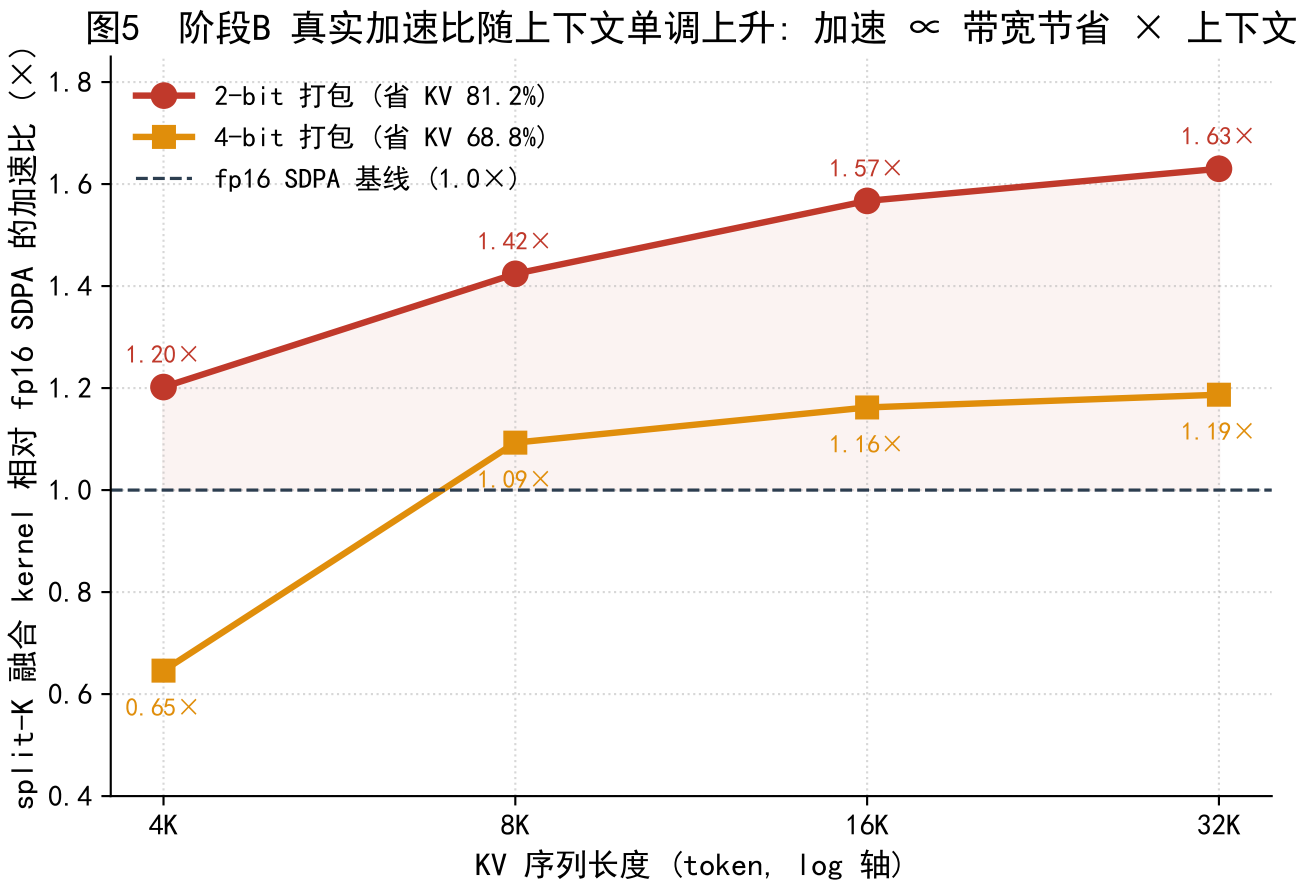


2-bit(省81%)加速 > 4-bit(省69%)，两档均随上下文单调上升 → 直接印证 decode 带宽受限。口径：单层/单query步 microbench(warmup25/中位数100/误差~1e-5)，非端到端。来源：results/kernel_decode_bench.json